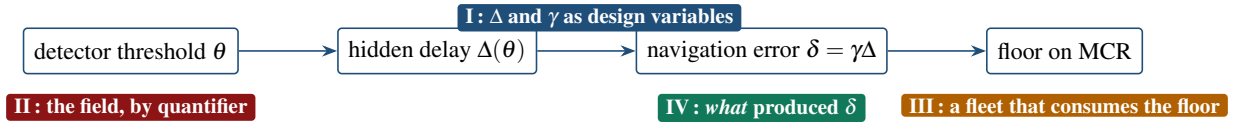


Four Follow-on Papers from *From Wire to Flight*

Research programme brief for supervisors · Roger Nick Anaedevha · RobustUAVs.ai · 1 October 2026

Parent manuscript: *From Wire to Flight: A Conditional Mission-Completion Guarantee for UAVs* (with M. Conti and F. Corò; targeted at IEEE S&P 2027). Papers I–II grow from its stated limitations; Papers III–IV from the CERTIFLIGHT MSCA proposal.

One chain, four papers. The parent paper proves a chain from a network detector's threshold θ to a certified floor on the mission-completion rate. Each follow-on paper attacks a different link, so the four do not overlap and each cites the others.



Paper I What Authenticity Costs in Metres: A Certified Budget for Cryptography and Sensing in UAV Navigation

Venue: IEEE S&P, main track · **Source:** merges proposals P1 (alternative PNT) and P2 (post-quantum signing) · **Effort:** ≈ 5 months

Core innovation Authentication appears on *both* sides of the certified budget $\rho = (\Delta + \Delta_{\text{auth}}) \gamma e^{LT_c} \leq m$. Authenticating a navigation source lowers γ (it can be trusted); authenticating a message raises Δ (signing takes time). The certificate decides the net sign:

$$\text{authentication pays} \iff \Delta_{\text{auth}} < \Delta(\theta) \left(\frac{\gamma_a}{\gamma_a} - 1 \right).$$

Two further results: a tighter detector makes authentication *more affordable but less beneficial*; and signatures protect against delay only via a freshness check whose clock is itself GNSS-disciplined.

Approach Measure γ for three navigation sources (GNSS done; inertial-only from flight logs already on disk; visual-inertial). Measure the cost of five schemes (HMAC-SHA256, Ed25519, ML-DSA-44, SLH-DSA, TESLA) on an STM32H7-class board and a companion computer, over classic CAN, CAN-FD and MAVLink 2; validate fragmentation against real bus timing (HCRL UAVCAN). Map the crossover per (transport, scheme, source).

Expected results From the parent's constants [DERIVED]: an authentication budget of 0.445 s at $m=5$ m, 1.39 s at 10 m, and *none* at 2 m. Prediction [ESTIMATE]: ML-DSA-44 costs ≈ 45 ms per message — affordable — yet saturates a classic CAN bus near 22 msg/s. *The cost is a cliff, not a slope.*

Status: budget arithmetic done; inertial $\gamma \approx 1$ week. **Main risk:** hardware access (fallback: pinned ARM, tagged as such).

Paper II SoK: The Quantifier Gap Between Certified Robustness and Airworthiness Assurance

Venue: IEEE S&P, SoK track (fallback: DASC / ACM CSUR) · **Source:** broadens P3 · **Effort:** ≈ 3 months, no measurement

Core innovation Certified guarantees and aviation assurance objectives both carry a *quantifier*: per input, per trajectory, per operation, or per population. Evidence transfers only when the guarantee's quantifier covers the objective's, or through a *lift* (a dynamics bound, a concentration argument, a runtime monitor) whose assumption is itself an assurance obligation. The parent paper is a worked instance: three lifts, three stated conditions.

Approach Systematic review 2017–2026 across security, ML, verification and avionics venues; stated inclusion criteria; snowballing to saturation; two independent coders with Cohen's κ . Code SORA 2.5, EASA learning-assurance and ED-324 objectives on the same fields; join the two into a transfer table.

Expected results The transfer table (discharge / evidence via lift / category error); an accounting of lifts as obligations; a ranked research agenda; a SORA Annex A evidence generator that keeps every assumption visible.

Status: framework defined; corpus *not yet collected* (expected 80–150 papers). **Main risk:** a thin or biased corpus.

Paper III Certificates as Constraints: Feasibility-Preserving Multi-Agent RL for Swarms Whose Guarantees Degrade

Venue: NeurIPS, main track · **Source:** CERTIFLIGHT O3, theorems T3–T4 (WP4) · **Effort:** ≈ 4 –5 months

Core innovation Safe RL offers constraints satisfied *in expectation* (Lagrangian, CPO) or *pointwise* from a hand-set margin (shields, barrier functions). Here the constraint set is the output of a formal certificate that *moves over time*: agent i may hold task τ only while its certified radius $R_i(t) \geq R_{\min}(\tau)$. Because qualifying radii are nested, feasibility and robustness collapse to linear-time checks: full coverage survives any k certificate collapses iff $n(\rho_j) \geq j + k$ for all j .

Approach A constrained MDP over the existing fleet simulator; a projection layer onto the certified-feasible set, active during training as well as deployment. Baselines: MAPPO, PPO-Lagrangian, CPO, shielded MADDPG, QMIX, and a fixed-margin allocator, each with and without the projection layer.

Expected results Zero pointwise violations *during training* (versus in-expectation satisfaction for Lagrangian baselines); a measured coverage-versus-degradation curve against the proved bound; an honest scalability ceiling.

Status: four propositions proved; the two combinatorial ones agree with a brute-force oracle on 3000/3000 random instances. RL study not yet run; simulator capped at 12 agents. **Reviewer question to pre-empt:** with an exact per-step check, learning must earn its place through travel time, switching cost and forecastable degradation — the study must beat a greedy oracle, not only RL baselines.

Paper IV When the Attacker Is Indistinguishable from the Weather: A Cross-Layer Benchmark and a Unified Perturbation Budget

Venue: USENIX Security · **Source:** CERTIFLIGHT O1–O2, theorems T1–T2 (WP2–WP3) · **Effort:** ≈5–6 months

Core innovation From the navigation input alone, shaped interference, a sensor fault and a gust are indistinguishable — so a certificate sound only against a declared attack model is unsound against an adversary who resembles the weather. The budget must therefore be source-agnostic: $\delta_{\text{tot}} = \delta_{\text{net}}(\theta) \oplus \delta_{\text{gnss}} \oplus \delta_{\text{fault}} \oplus \delta_{\text{env}}$, combining worst-case, stochastic and bounded-unknown sources in one certified floor.

Approach Add fault and environment terms to the certificate engine; characterise the union bound’s slack as a dependence-uncertainty (Makarov/VaR-aggregation) gap; release the cross-layer benchmark whose records carry an enforced provenance flag; check soundness at every operating point across seeds with Holm correction.

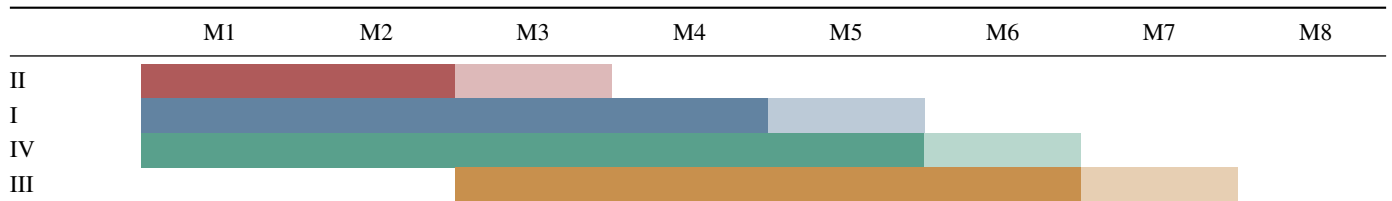
Expected results Evidence that the composed floor holds where a signature-tuned detector alone fails silently; a measured union-bound gap; a public benchmark positioned to become a community evaluation substrate.

Status: six-corpus ingest, certificate engine, residual-budget lemma, provenance flags in hand. **Open:** the Makarov reduction is *to be proved* (fallback: the sound union bound, gap reported empirically).

At a glance

Paper	Venue	Headline claim, one line	In hand	Biggest open item	Effort
I	S&P main	Authentication is both a credit and a debit on one certified budget; a closed-form crossover decides which wins.	derived budget; γ for GNSS; flight logs on disk	benchmark hardware	5 mo
II	S&P SoK	Guarantees and assurance objectives transfer only when their quantifiers match, or through a lift that is itself an obligation.	framework; coding scheme	literature corpus	3 mo
III	NeurIPS	A formal certificate can serve as a time-varying, pointwise constraint, with feasibility kept during learning.	four propositions; 3000/3000 oracle check	RL study; simulator >12 agents	4–5 mo
IV	USENIX Sec.	Because attacks, faults and gusts are indistinguishable at the input, only a source-agnostic budget is a security guarantee.	six-corpus ingest; engine; provenance flags	Makarov reduction; benchmark release	5–6 mo

Indicative timeline (months from approval; planning estimate, not a commitment)



Dark: research and experiments. Light: writing and submission. Paper II’s first two months are search and coding; Paper I front-loads the inertial-only γ (about one week, data already on disk); Paper III starts once I and IV are producing certified radii, though its theory is already done.

Proposed order

- II** first: cheapest, no hardware, and it fixes the vocabulary the other three cite.
- I** and **IV** in parallel: they share the certificate engine but need different hardware and data.
- III** last: it consumes the certified radii that I and IV produce.

Decisions requested from supervisors

- Approve the four-paper split and the venue for each.
- Paper I: access to a Pixhawk-class board and a CAN/CAN-FD interface, or agreement to the pinned-ARM fallback.
- Paper II: a second coder for the literature review (needed for Cohen’s κ).
- Paper III: GPU allocation on the group cluster, and agreement to extend the simulator beyond 12 agents.
- Lead and co-author arrangement for each paper.

What is established and what is not. Parent-paper values are *measured*: hidden delay saturates at 4.84 s over 15,392 relay hops; $\gamma \leq 1.625$ m/s over 674 post-onset samples, varying $6.1\times$ with conditions, calibrated in hover on one airframe. Paper I’s budget figures are [DERIVED] from those constants; its CAN-bus figures are an [ESTIMATE] from an assumed frame model, to be replaced by measurement. Paper III’s propositions are proved and checked against brute force; its learning results do not yet exist. Paper IV’s Makarov reduction is a claim to prove, with the sound union bound as fallback. Paper II’s corpus has not been collected. No item on these two papers is presented as a result before it exists.